Stanford MEDICINE Pediatrics
School of Medicine

Abstention and Threshold Identification for Uncertainty Management in Clinical Decision Tools: A Case Study using Human-In-The-Loop Pediatric Autism Classifiers

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INTRODUCTION

- The status quo of autism diagnosis is slow, subjective, and relies on outnumbered specialists, leading to delays that prevent critical early intervention
- Previous work has demonstrated that ML classifiers can accurately diagnose autism using 30 behavioral features from home videos with features rated by non-specialists
- In this work, we implement an abstention mechanism through thresholding to make diagnosis predictions more clinically interpretable and trustworthy
- We evaluate this thresholding approach on a large, noisy dataset to examine the trade-offs between diagnostic performance and retained sample predictive coverage

VIDEO DATASET

- Curated a purposefully noisy and heterogeneous dataset to model a scenario where abstention may be required often
- >1500 videos sourced from parent submissions, YouTube, GuessWhat mobile application, and other past studies
- Classified videos through clinician input or parent-sourced data as children with autism and without autism
- Diagnostic feature ratings were done by 56 clinicians and 50 non-clinicians, resulting in 8,144 sets of ratings across 1031 unique children (588 with ASD, 443 without)

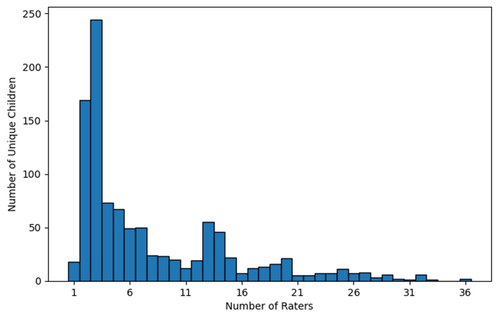

Fig. 2: Frequency distribution of the number of raters per child, where the majority of children were given at least three sets of ratings.

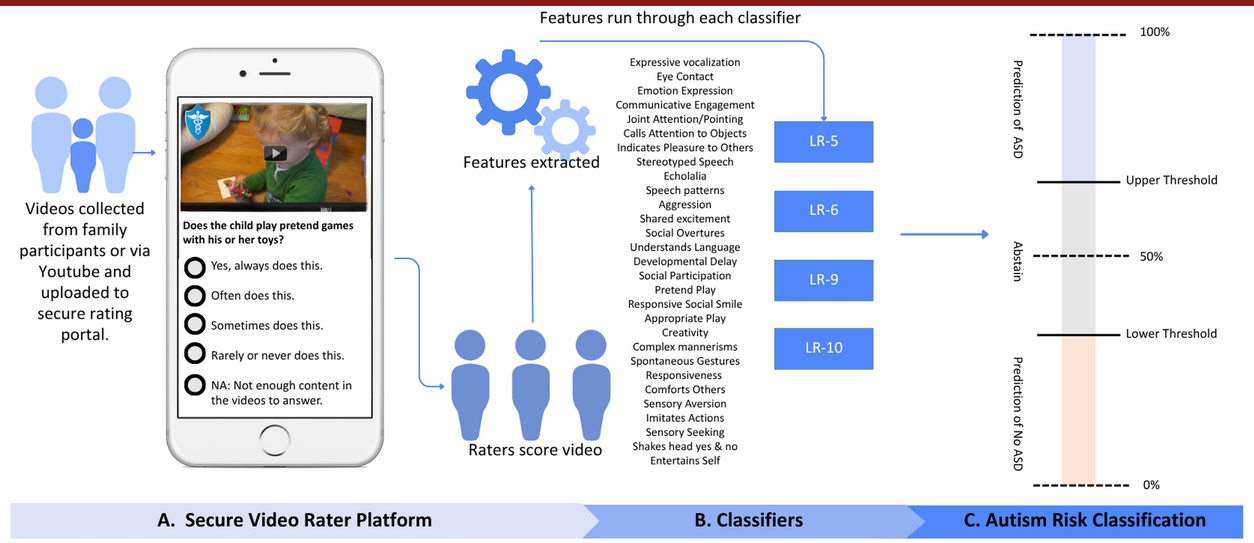

Fig. 1: (A) Short home videos of children were rated by human labelers in a secure rating platform, where 30 behavioral features were rated on a 4-point scale. (B) These features were used in the following analysis by our previously developed machine learning classifiers to determine an autism risk classification. (C) The current study examines how we can determine and utilize thresholding values to tell the machine learning models to abstain when a certain level of uncertainty is reached.

METHODS & RESULTS

Thresholding experiments:

- Leveraged logistic regression classifiers trained on autism instrument data from past work (LR5, LR6, LR9, LR10) and outputted inferences for each unique child in video ratings
- Replaced the standard single 0.5 decision threshold for LR probability outputs with lower and upper thresholds (t_L, t_U) to create an abstention window for uncertain cases
- Conducted a grid search on millions of threshold combinations to evaluate the full range of possible outcomes

Baseline classification:

- Performance from baseline 0.5 single-decision threshold was varied, but performed within reason given context of the noisy dataset

Model	Accuracy	Sensitivity	Specificity	PPV	NPV
LR5	62.08%	56.12%	69.98%	71.27%	54.58%
LR6	68.57%	66.50%	71.33%	75.48%	61.60%
LR9	71.87%	82.31%	58.01%	72.24%	71.19%
LR10	66.92%	64.63%	69.98%	74.07%	59.85%

Table 1: Baseline performance of logistic regression binary classifiers on the curated collection of 1031 naturalistic video samples.

Thresholding performance:

- Evaluated LR6 due to greatest equivalence between sensitivity & specificity to study impact of uncertainty management on performance
- Chose specific ceilings for abstention rate and searched for the threshold pairs with maximum performance satisfying each ceiling (such as with Youden's J, or sensitivity + specificity - 1)

Abstain	Accuracy	Sensitivity	Specificity	Youden's J	t_L	t_U
Baseline	68.57	66.50	71.33	.3783	N/A	N/A
0.1	71.77	71.37	72.28	.4365	0.4027	0.5480
0.2	73.58	69.89	78.33	.4822	0.3685	0.6923
0.3	76.31	74.45	78.78	.5323	0.2966	0.7530
0.4	79.64	85.87	70.52	.5639	0.1944	0.7060
0.5	81.59	86.38	74.88	.6126	0.1645	0.8459

Table 2: LR6 model performance with optimized thresholds (t_L, t_U) under varying abstention ceilings, searching for the highest Youden's J.

- Increasing the maximum tolerated abstention rate allows the model to achieve a significantly higher optimal performance
- Demonstrates clinical viability through significant performance boost at tolerable abstention (e.g. 30%)

METHODS & RESULTS (cont.)

Performance-Abstention Trade-offs:

- We also examine trends over ranges of performance targets, setting combinations of PPV/NPV and sensitivity/specificity targets and measuring minimum abstention rate to achieve said targets

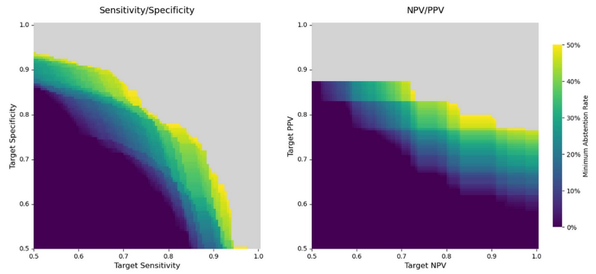

Fig. 3: Performance trade-off landscapes for the LR6 model, showing achievable targets at an abstention rate of 50% or less. The grey area represents target combinations that are either impossible to achieve or require an abstention rate greater than 50%.

- Clinicians can set specific performance targets and output a threshold pair alongside a minimum abstention rate (e.g. NPV=0.95, PPV=0.75, abstain<0.2) and confirm if viable for their usage
- Could use this information to confidently identify negative cases while flagging a smaller, manageable subset of potential positive cases for further review

CONCLUSIONS, LIMITATIONS AND FUTURE DIRECTIONS

- Demonstrated clinical viability of using thresholding to flexibly manage uncertainty for AI-based diagnosis tools
- Can be utilized in many different contexts on an ad hoc basis, such as a triage tool to easily rule out negative cases or a 'first pass' model with high abstention tolerance but confident decisions on non-abstained cases
- Limitations include dataset size, incomplete metadata, and unquantified rater variability
- Model calibration and use of a dedicated evaluation dataset pose critical steps for future clinical use
- Future work could train models directly on video ratings rather than instrument data, exploring uncertainty management prior to post-inferencing stages

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